

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1: Experiments

**COURSE NAME:** Cloud Computing and  
Big Data Analytics

**COURSE CODE:** CSA1596

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### COURSE OUTCOME COVERED

**CO3:** *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)  
Question Count: 5

**Total Marks: 40**

Duration: 60 Minutes

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### Code of Conduct

I M. Pranay (192525382)\_certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Pranay\_

### Experiments Rubric

| Criteria                 | Weightage | Level 4 - Exemplary                                   | Level 3 - Proficient              | Level 2 - Developing   | Level 1 - Beginning             |
|--------------------------|-----------|-------------------------------------------------------|-----------------------------------|------------------------|---------------------------------|
| Experimental Design      | 25%       | Design is systematic and technically strong           | Mostly correct design             | Basic design with gaps | Poorly designed activity        |
| Use of Tools / Platforms | 20%       | Appropriate and effective use of cloud tools          | Good tool usage with minor issues | Limited tool usage     | Incorrect or missing tool usage |
| Application of Concepts  | 20%       | Concepts are applied accurately to practical tasks    | Mostly accurate application       | Partial application    | Weak application                |
| Analysis and Output      | 20%       | Strong analysis and meaningful outcome interpretation | Good analysis with minor gaps     | Basic analysis         | Little or no analysis           |
| Documentation            | 15%       | Clear, structured, and complete documentation         | Mostly complete documentation     | Partially complete     | Poor documentation              |

Total Marks Awarded:

## Assessment Tasks-I

### Experiments

**Experiment 1:** Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

**Experiment 2:** Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org, over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

**Experiment 3:** Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

**Experiment 4:** Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

## Assessment Tasks-II

### Experiments

**Experiment 1:** Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

**Experiment 2:** Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

**Experiment 3:** Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

**Experiment 4:** Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

**Experiment 1:** Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

**Experiment 2:** Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement, Rental Agreement, Area, place, Range, property value etc.

**Experiment 3:** Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

**Experiment 4:** Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

# Assessment Tool 1

## Experiment 1: Library book reservation system

SIMATS Library Book Reservation...

SIMATS Library

SIMATS Library

SIMATS Library Report

SIMATS Library Booking System

M

MANEDI PRANAY

192525382.simats@saveetha.com

My Account

Home

Sign out

M MANEDI PRA...

SIMATS Library

Student Name \*  
First Name Last Name

Register Number \*  
#####

Book ID \*

Book Title \*

Author \*

Publication \*

Published Year \*  
#####

Edition \*

Number of Copies \*  
#####

Rack Number \*

Department \*  
-Select-

Book Category \*  
-Select-

Booking Status \*  
-Select-

SIMATS Library Book Reservation...

SIMATS Library

SIMATS Library

SIMATS Library Report

SIMATS Library Booking System

M

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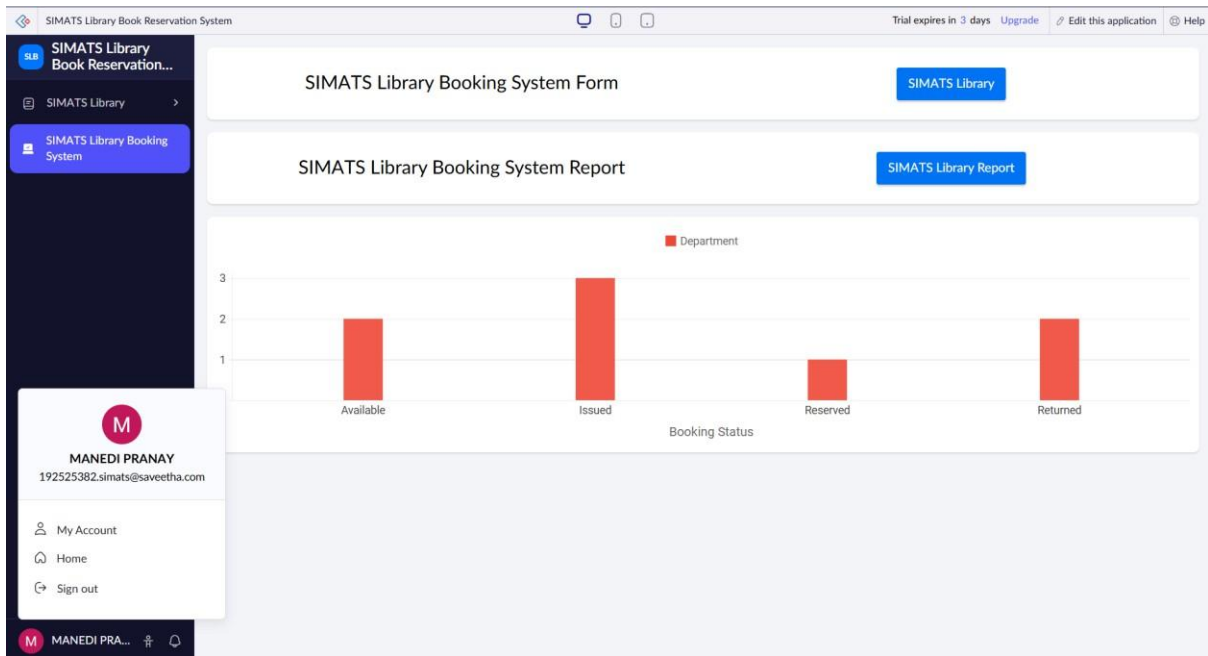
M MANEDI PRA...

SIMATS Library Report

Author Book Title Book ID Publication Published Year Edition Number of Copies

|                 |                 |         |             |      |      |    |
|-----------------|-----------------|---------|-------------|------|------|----|
| Rajkumar        | Cloud Computing | B008    | Elsevier    | 2019 | 2nd  | 2  |
| Stuart Russell  | AI              | B007    | Pearson     | 2026 | 4th  | 4  |
| Abraham         | Database System | B006    | Graw Hill   | 2025 | 11th | 9  |
| Silberschatz    | CN              | B005    | Wiley       | 2024 | 10th | 7  |
| Seymour         | Python          | B004    | McGraw      | 2020 | 6th  | 5  |
| Herbert Schildt | Java            | B002    | McGraw-Hill | 2021 | 12th | 8  |
| Reema Thareja   | OS              | B001    | Oxford      | 2022 | 3rd  | 10 |
| John            | CV              | 2165616 | sensor      | 2023 | 2026 | 1  |

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## Experiment 2: Payroll Processing System

The screenshot displays the Payroll Processing System dashboard. The sidebar on the left contains navigation links for 'Payroll Processing System', 'Pay Master', 'Pay Master Report', and 'Payroll Processing System'. The main content area features a 'Pay Master' form with various input fields for employee information and salary details. The user profile dropdown for 'MANEDI PRANAY' is also visible.

| Field            | Value    |
|------------------|----------|
| Employee ID *    |          |
| Employee Name *  |          |
| Department *     | -Select- |
| Designation      |          |
| Basic Salary     | *****    |
| HRA              | *****    |
| DA               | *****    |
| Other allowances | *****    |
| Gross Salary     | *****    |
| PF Deduction     | *****    |
| Tax Deduction    | *****    |
| Other Deduction  | *****    |
| Net Salary *     | *****    |

Payroll Processing System

Pay Master

Pay Master Report

Payroll Processing System

MANEDI PRANAY

192525382.simats@saveetha.com

My Account

Home

Sign out

Pay Master Report

| Employee ID | Employee Name | Department | Designation          | Basic Salary | HRA    | DA      |
|-------------|---------------|------------|----------------------|--------------|--------|---------|
| EMP008      | Rohith        | Sales      | Java+                | 9546456      | 684684 | 46846   |
| EMP007      | Giri          | Sales      | c+                   | 45965        | 654    | 6846    |
| EMP006      | Sai           | Marketing  | c++                  | 94631        | 65465  | 6455322 |
| EMP005      | Pranay        | Sales      | Java                 | 4798         | 8646   | 854563  |
| EMP004      | SREEKAR       | IT         | AI                   | 35468        | 554    | 5451    |
| EMP003      | Ariya         | HR         | Accountant           | 45000        | 9000   | 4500    |
| EMP002      | Prince        | Finance    | HR                   | 5000         | 1000   | 500     |
| EMP001      | Sai Rahul     | HR         | Software Engineering | 50000        | 10000  | 5000    |

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Payroll Processing System

Pay Master

Payroll Processing System

MANEDI PRANAY

192525382.simats@saveetha.com

My Account

Home

Sign out

Pay Master Form

Pay Master Report

Department

|           |   |
|-----------|---|
| Finance   | 1 |
| HR        | 2 |
| IT        | 1 |
| Marketing | 1 |
| Sales     | 3 |

https://creatorapp.zoho.in/192525382.simats\_saveetha/payroll-processing-system

## Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor

Home

SIMATS LAB VM

 SIMATS LAB VM

 Resume this virtual machine

 Edit virtual machine settings

▼ Devices

 Memory

4 GB

 Processors

2

 Hard Disk (SCSI)

15 GB

 CD/DVD (SATA)

Using file C:\Pro...

 Network Adapter

NAT

 USB Controller

Present

 Sound Card

Auto detect

 Display

Auto detect

▼ Description

Type here to enter a description of this virtual machine.



```
Network boot from Intel E1000
Copyright (C) 2003-2021 VMware, Inc.
Copyright (C) 1997-2000 Intel Corporation
CLIENT MAC ADDR: 00 0C 29 C5 78 21 GUID: 564D96B4-BAF3-385E-E196-B7C1EDC57821
PXE-E53: No boot filename received
PXE-MBF: Exiting Intel PXE ROM.
Operating System not found
```

▼ Virtual Machine Details

**State:** Suspended

**Snapshot:** 11-08-2026 11:12:56

**Configuration file:** C:\Users\Amrutha\Documents\Virtual Machines\SIMATS LAB VM\SIMATS LAB VM.vmx

**Hardware compatibility:** Workstation 17.5 or later virtual machine

**Primary IP address:** Network information is not available



# Experiment 4:

## Ubuntu 64-bit (4)

- ▶ Resume this virtual machine
- 🔧 Edit virtual machine settings

▼ Devices

|                  |                      |
|------------------|----------------------|
| Memory           | 2 GB                 |
| Processors       | 1                    |
| Hard Disk (SCSI) | 15 GB                |
| CD/DVD (SATA)    | Using file C:\Pro... |
| Network Adapter  | NAT                  |
| USB Controller   | Present              |
| Sound Card       | Auto detect          |
| Display          | Auto detect          |

▼ Description

Type here to enter a description of this virtual machine.

```
Network boot from Intel E1000
Copyright (C) 2003-2021 VMware, Inc.
Copyright (C) 1997-2000 Intel Corporation

CLIENT MAC ADDR: 00 0C 29 AA 4E 33  GUID: 564D60A8-4401-EA30-251B-
PXE-E53: No boot filename received

PXE-M0F: Exiting Intel PXE ROM.
Operating System not found
```

▼ Virtual Machine Details

**State:** Suspended  
**Snapshot:** Snapshot 1  
**Configuration file:** C:\Users\Amrutha\Documents\Virtual Machines\Ubuntu 64-bit (4)\Ubuntu 64-bit (4).vmx